
accelerate
collaborate
innovate
communicate
palm
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Wireless Enabling Technologies for Palm

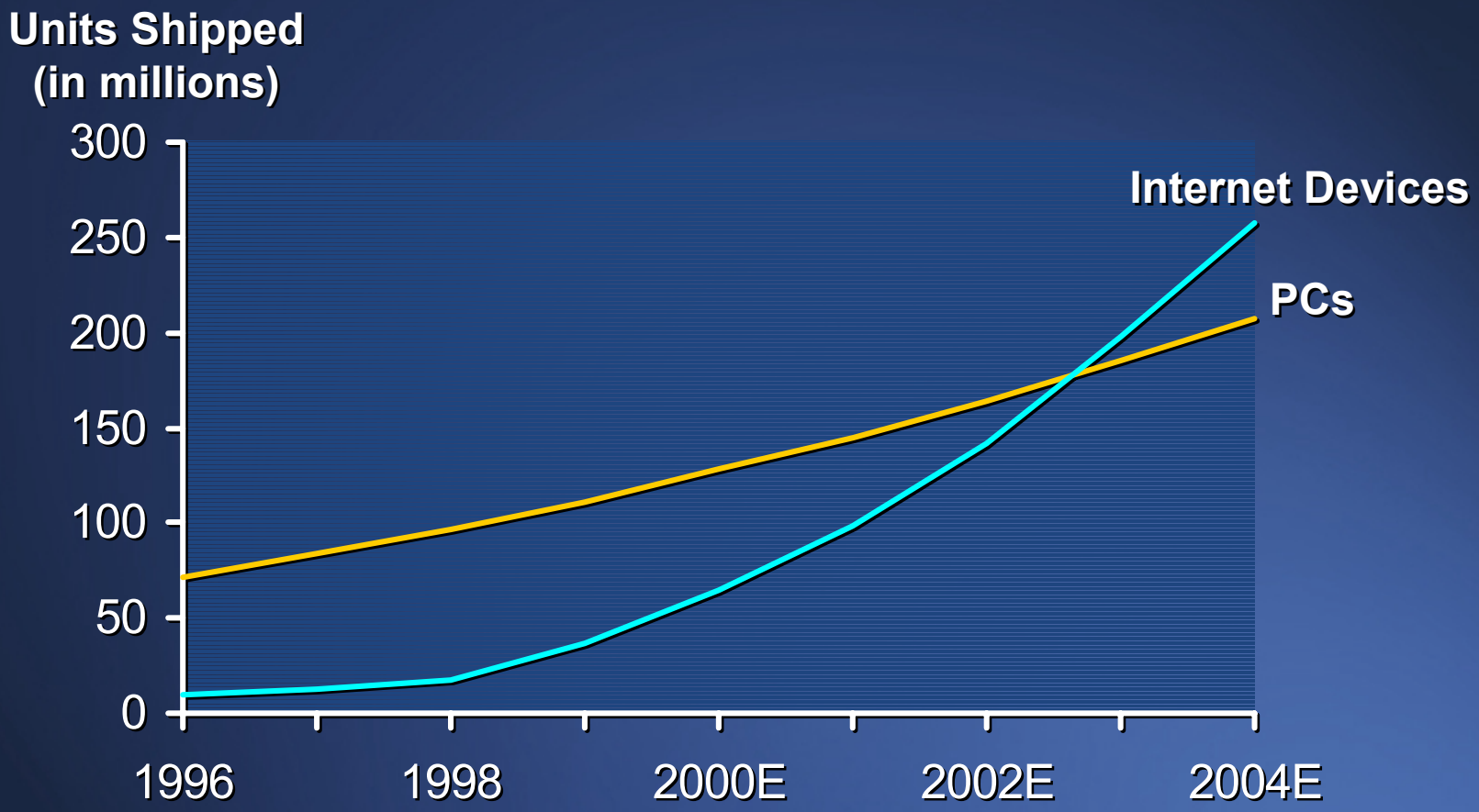
Steve Butsch
Director, Systems Engineering

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AGENDA

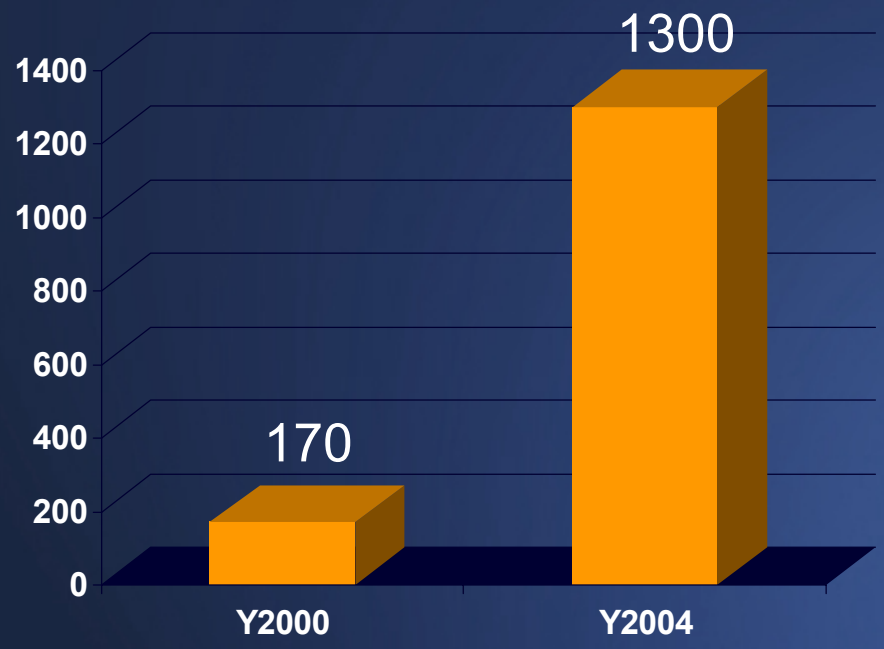
- Wireless—becoming pervasive
- Palm as wireless terminal
- Corporate LAN to wireless WAN Connectivity
- Building wireless applications—Web
- Building wireless applications—Client/Server
- Third-party hosting/WASPs
- Software tools

Devices and Content Are Shifting Away From the PC

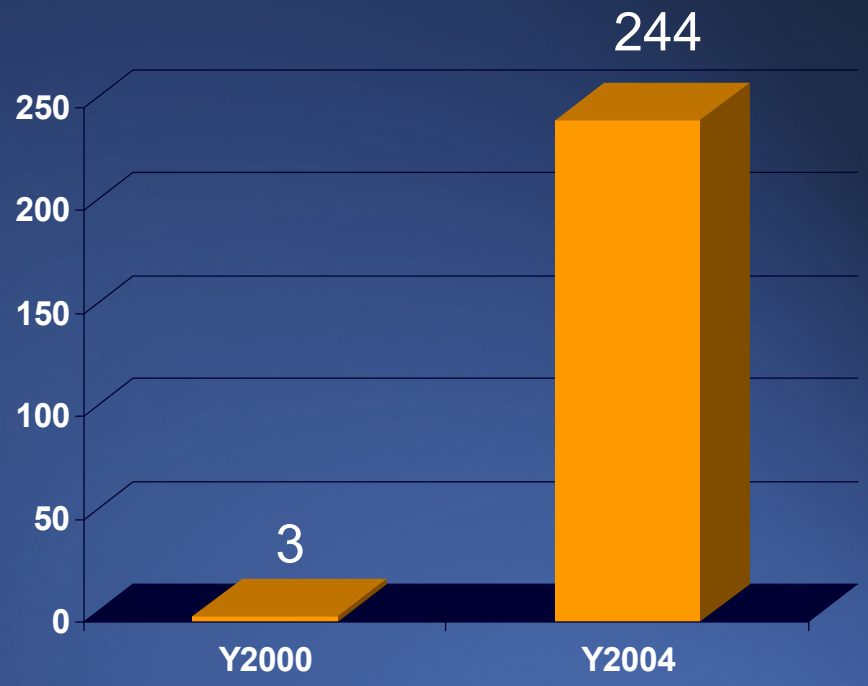


Source: Morgan Stanley Dean Witter Equity Research, IDC, The Yankee Group

Wireless Projections



Data Subscribers
(millions)



Messages per Month
(billions)

Source: Cahners In-Stat

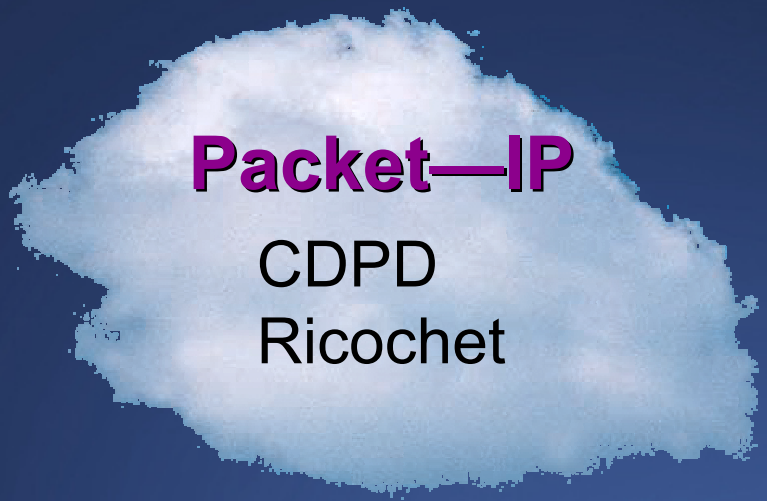
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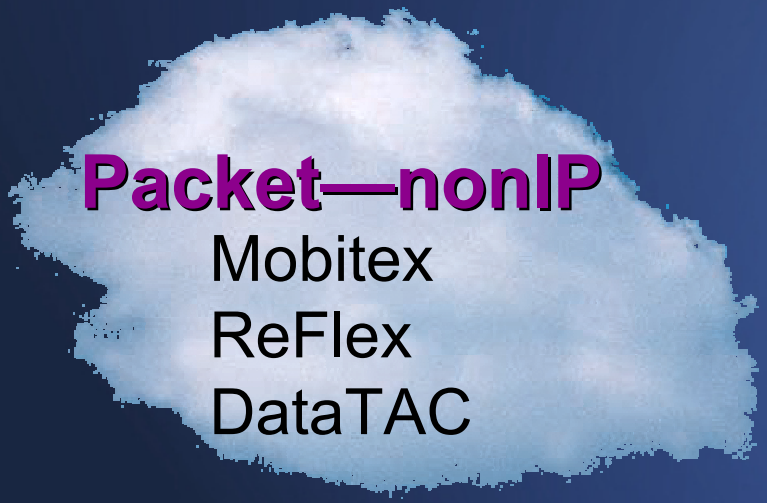
Wireless Network Survey

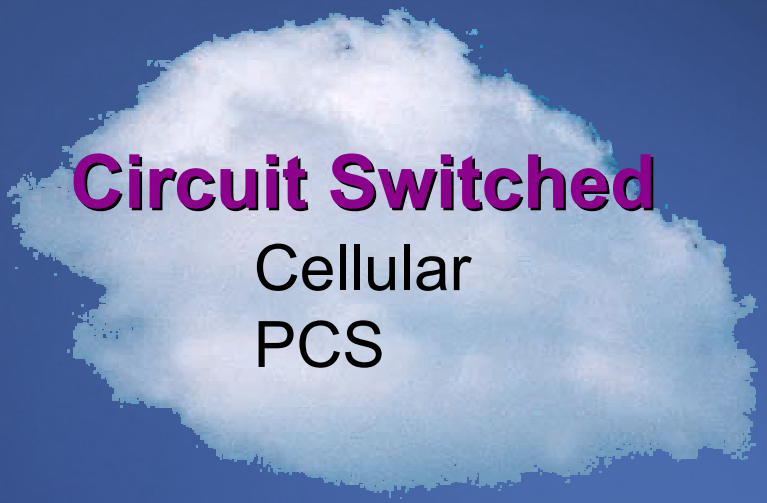
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U.S. Wireless Networks



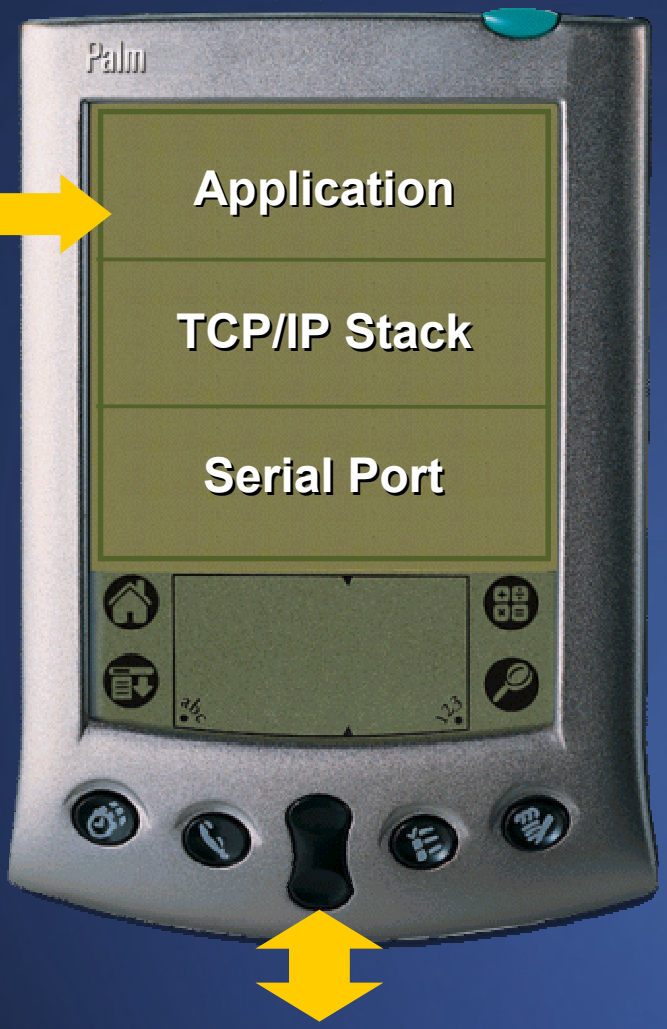




IP vs. nonIP

TCP/IP Clients
HSS Client
AvantGo Client
Puma IA Client
MultiMail Client
XTND/IBM Client
OmniSky Client
Etc.

Palm III,V, VII Series



Palm VII Series



Ethernet Cradle, Telephone Modem
Cell Phone, CDPD Modem

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Mobitex

- BellSouth Wireless Data
- Nationwide network
- Mobitex Technology
  - Cell based
  - Packet switched, (MPAK, not IP)
  - 800–900 MHz, 8 KBPS
- Coverage in the top 266 MSAs
  - Translates into top 7800 cities and towns
  - 93% of the US business population

Advantages of Wireless Packet Data

- Better response time for short messaging
- Reliable
- Virtually connected
- Tolerant of coverage gaps
- Spectrum efficient

ReFLEX™—Two-Way Paging

- Skytel SkyWriter Services
  - Other players: Airtouch paging, Pagnet, Pagemart, Metrocall
- Based on Narrowband PCS
  - Licensed 50 KHz blocks in 900 MHz spectrum
- Asymmetric transmit/receive channels
  - Outbound 1.6 to 6.4 KBPS
  - Inbound 0.8 to 9.6 KBPS

CDPD: Cellular Digital Packet Data

- Offered by AT&T, Verizon, Ameritech, Southwestern Bell, US Cellular and others
- Regionally based w/ roaming agreements
  - Coverage in top 120 MSAs
- CDPD Technology
  - Overlay on existing analog cellular (AMPS)
  - Packet switched—TCP/IP
  - 800–900 MHz, 19.2 KBPS
  - TCP/IP based

Ricochet

- Metricom Inc.
- Wide-area, spread-spectrum network
  - Unlicensed frequency band (902–928 MHz)
  - Packet switched, TCP/IP
- Coverage
  - 28.8 KBPS: Seattle, Wash DC
  - 128 KBPS: Atlanta, Baltimore, Dallas/Ft Worth, Houston, Philadelphia, Phoenix, San Diego, and San Francisco Bay Area
    - *Rolling out: Boston, Chicago, Denver, Detroit, Kansas City, Los Angeles, New York City, Miami, Minneapolis, Salt Lake City, St. Louis, Seattle, and Washington, D.C.*

Cellular/PCS

Technology	Classification	Frequency Band
AMPS	Analog Cellular	800 MHz
CDMA (IS-95)	Digital Cellular or PCS	800 MHz, 1900 MHz
TDMA (IS-136)	Digital Cellular or PCS	800 MHz, 1900 MHz
GSM 1900	PCS	1900 MHz

Wireless Future—2.5G

- 2.5G takes advantage of existing infrastructure and frequencies—typically 115kbps to 307kbps
- Deployments—2000–2002
  - GPRS
    - Packet services on GSM
    - North America and Europe
  - CDMA2000 1X or 1XRTT
    - Packet services on CDMA
    - North America
  - EDGE
    - Packet services for TDMA and GSM
    - North America

Wireless Future—3G

- 3G requires new frequencies (licenses) and new infrastructure
- Deployments: 2002–2003
  - WCDMA
    - Likely to become global standard
    - Data rates up to 2MBPS
  - CDMA2000
    - North American focused
    - Data rates up to 2MPBS
- ITU has decreed that 3G wireless devices must support WCDMA and CDMA2000



Palm™ as Wireless Terminal

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PalmVII/VIIx

- BSWD Network
- A Palm Computing® platform device
- Instant access to Internet data
- Wireless Internet messaging



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Minstrel Wireless IP Modem

- Novatel Wireless Inc.
- Compatible Palm III and V series
- CDPD wireless technology
- Compatible with most ISPs



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Qualcomm pdQ Phone

- CDMA technology
  - Up to 14.4 Kbps
- Palm OS compatible
  - 2 MB memory
  - 160x240 display
- Web, Email and Paging
  - pdQ 800
  - pdQ 1900



Digital Phone Support

- Over 50 models supported
  - Nokia
  - Ericsson
  - Motorola
  - Kyocera
  - Others
- Cable
- IR







Palm to Ricochet

- Ricochet SE Modem and cable
- Enables all Palm IP based applications
  - Email
  - Web
  - Server sync



Symbol 1700 Series

- 1D and 2D bar code scanning
- IEEE 802.11 wireless LAN
- Future: GSM and CDPD support



Handspring VisorPhone

- GSM Springboard module
- Caller ID, Call waiting
- 3-way calling
- Speed dial
- Call history (stores up to 1,000 messages)
- SMS text messaging



Palm "Mobile Internet Kit"

- Contents
  - Palm OS 3.5 Upgrade
  - Web Clipping
  - E-Mail (POP3, IMAP4)
  - SMS
  - WAP



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Palm's Latest Initiatives in Voice and Data



- Motorola tri-band GSM phone with Palm OS: 2002



- RealVision GSM snap on for PalmV/VX: Early 2001



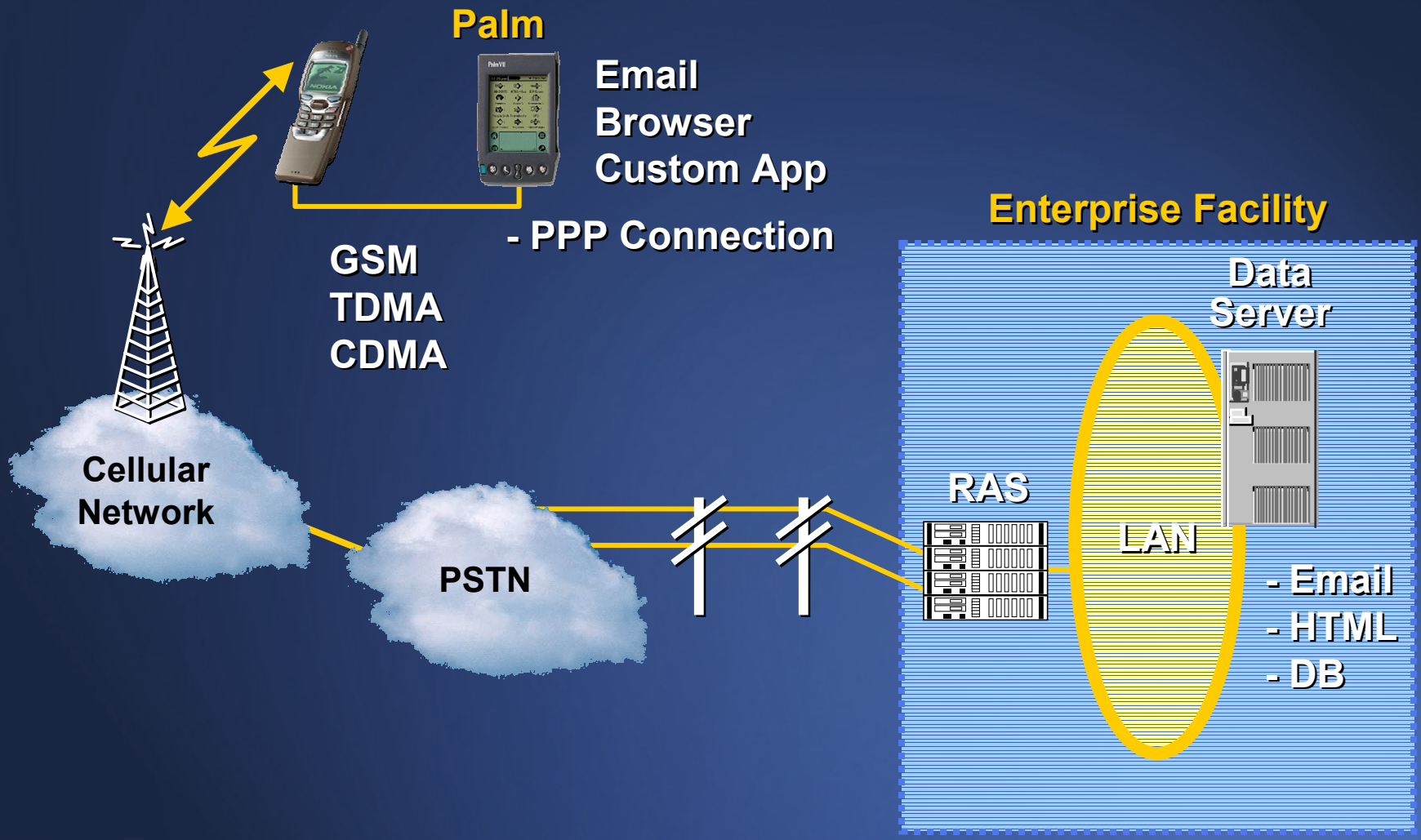
- Kyocera smart phone with Palm OS: Late 2000

Corporate LAN to Wireless WAN Connectivity

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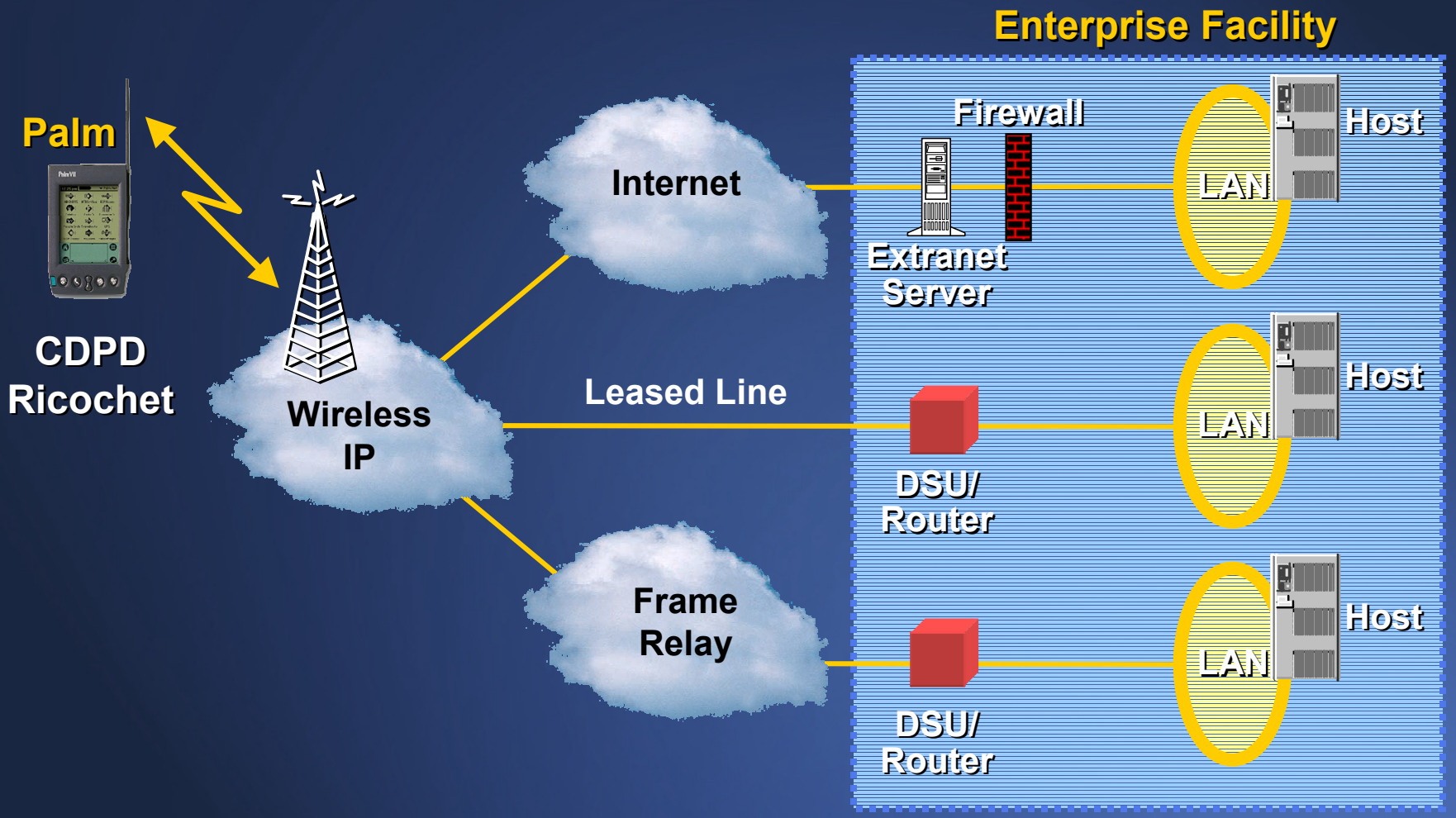
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Connecting Corporate LAN to Wireless—Cellular



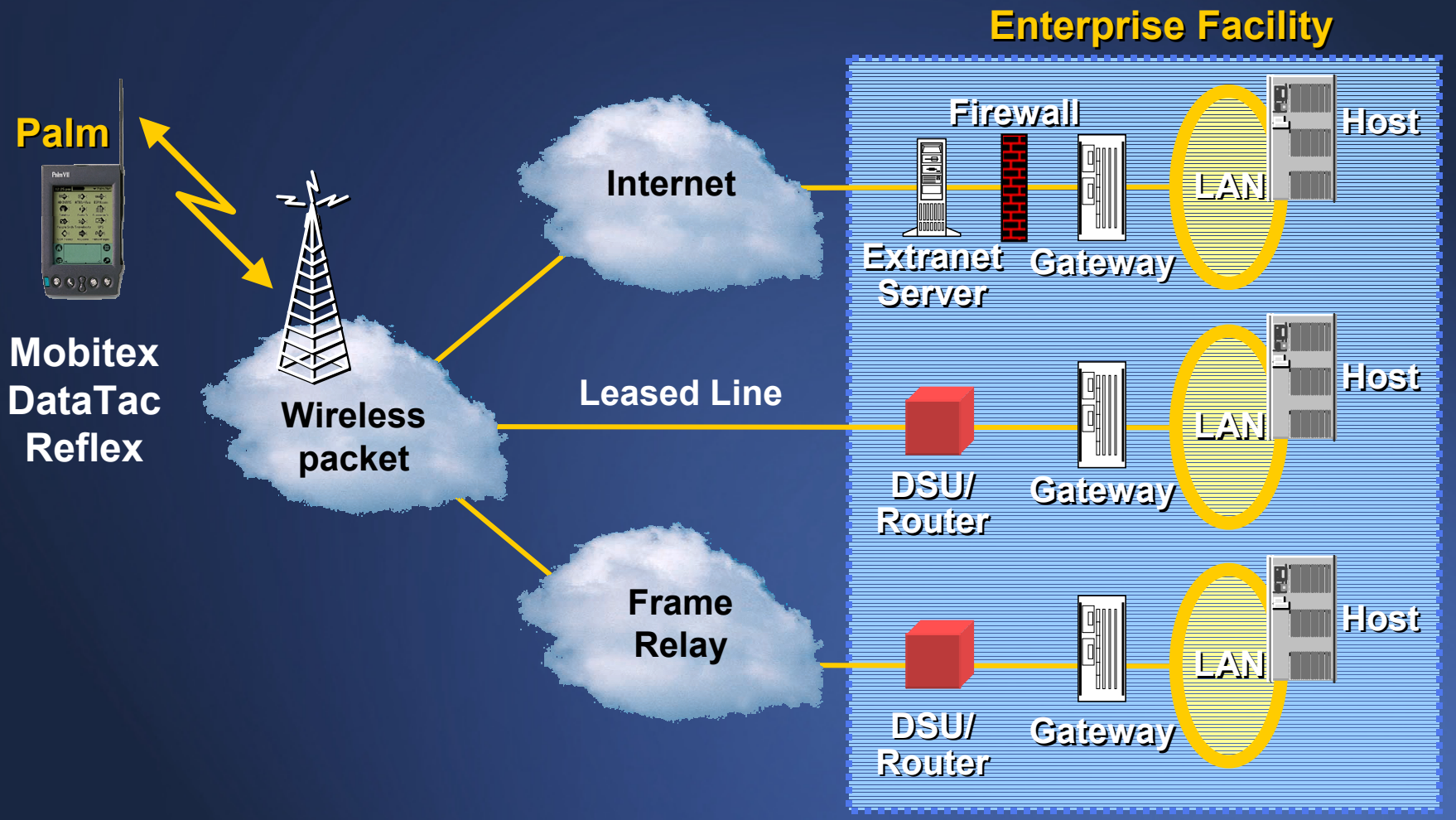
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Connecting Corporate LAN to Wireless—IP Packet



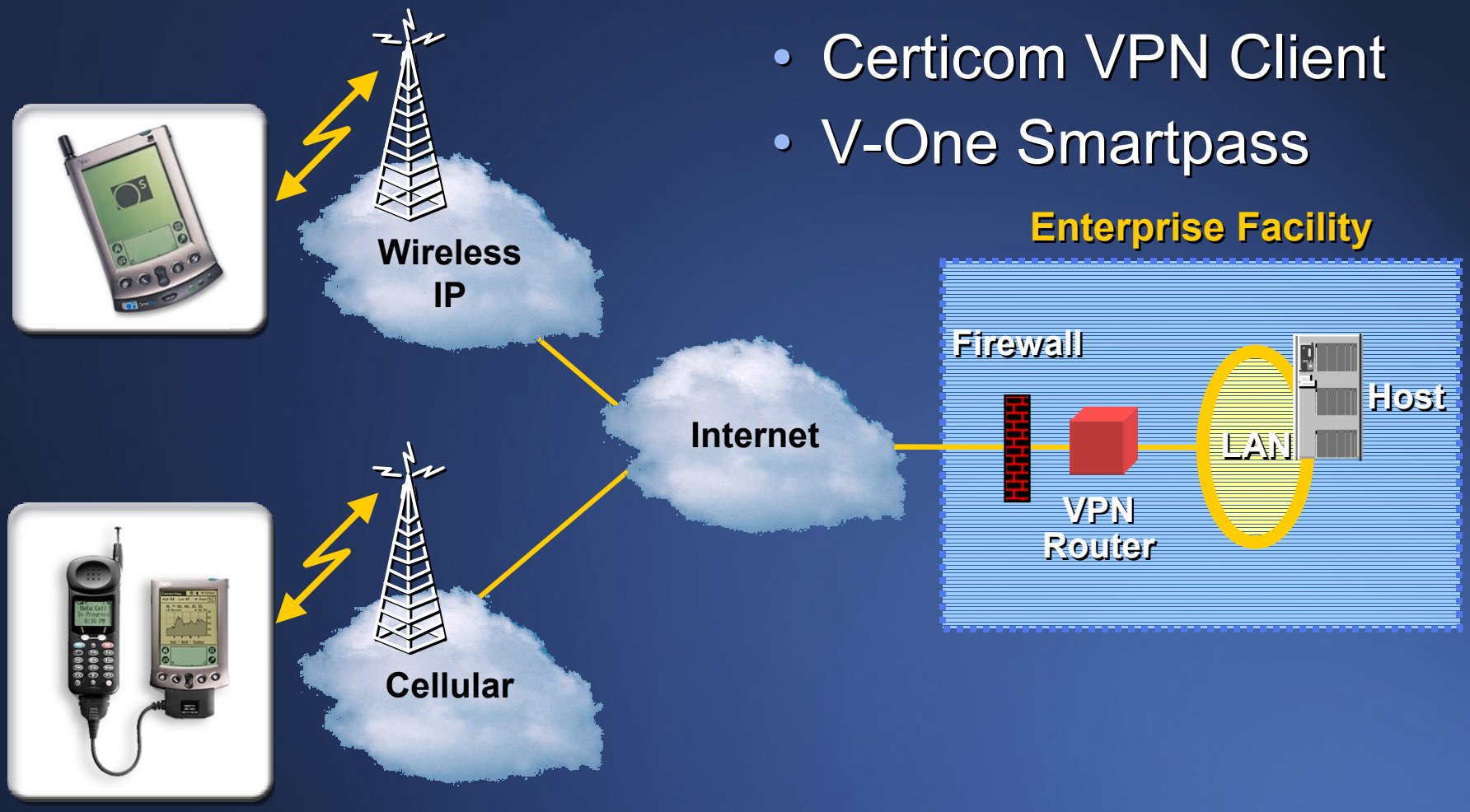
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Connecting Corporate LAN to Wireless—Packet

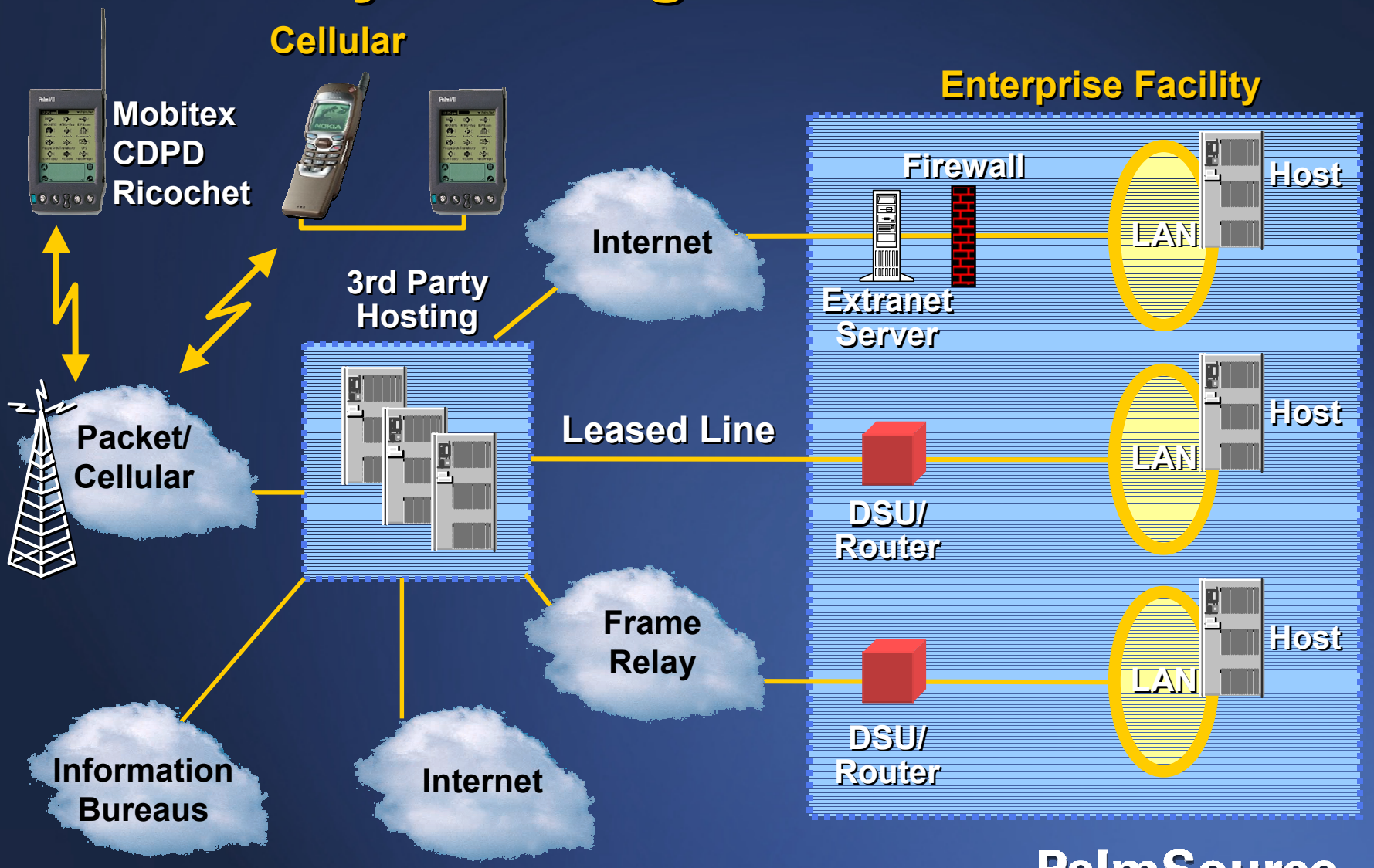


Palm VPN Clients—In Beta

- Certicom VPN Client
- V-One Smartpass



3rd Party Hosting



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Building Wireless Applications —Web-Based Technologies

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Why Wireless Web?

- Lots of content, both public and corporate, is stored as markup language
- Easy to program with markup language
- Request/Response protocol is well suited for narrowband wireless applications

Markup Languages

- HTML—HyperText Markup Language
  - The lingua franca of the Internet
- HDML—Handheld Device Markup Language
  - Developed by Phone.com
  - Used by major of U.S. browser enable phones
  - Not compatible with WML
- WML—Wireless Markup Language
  - Introduced by WAP forum
  - More feature rich than HDML (i.e., better scripting)
- xHTML—extensible HTML
  - Convergence of WML and I-Mode?

XML Primer

Car for sale
Model : Pontiac Firebird, Build in 1993.
Red color and in good condition. 150 horsepower.
Contact : Rob Janson, 425 462 1999

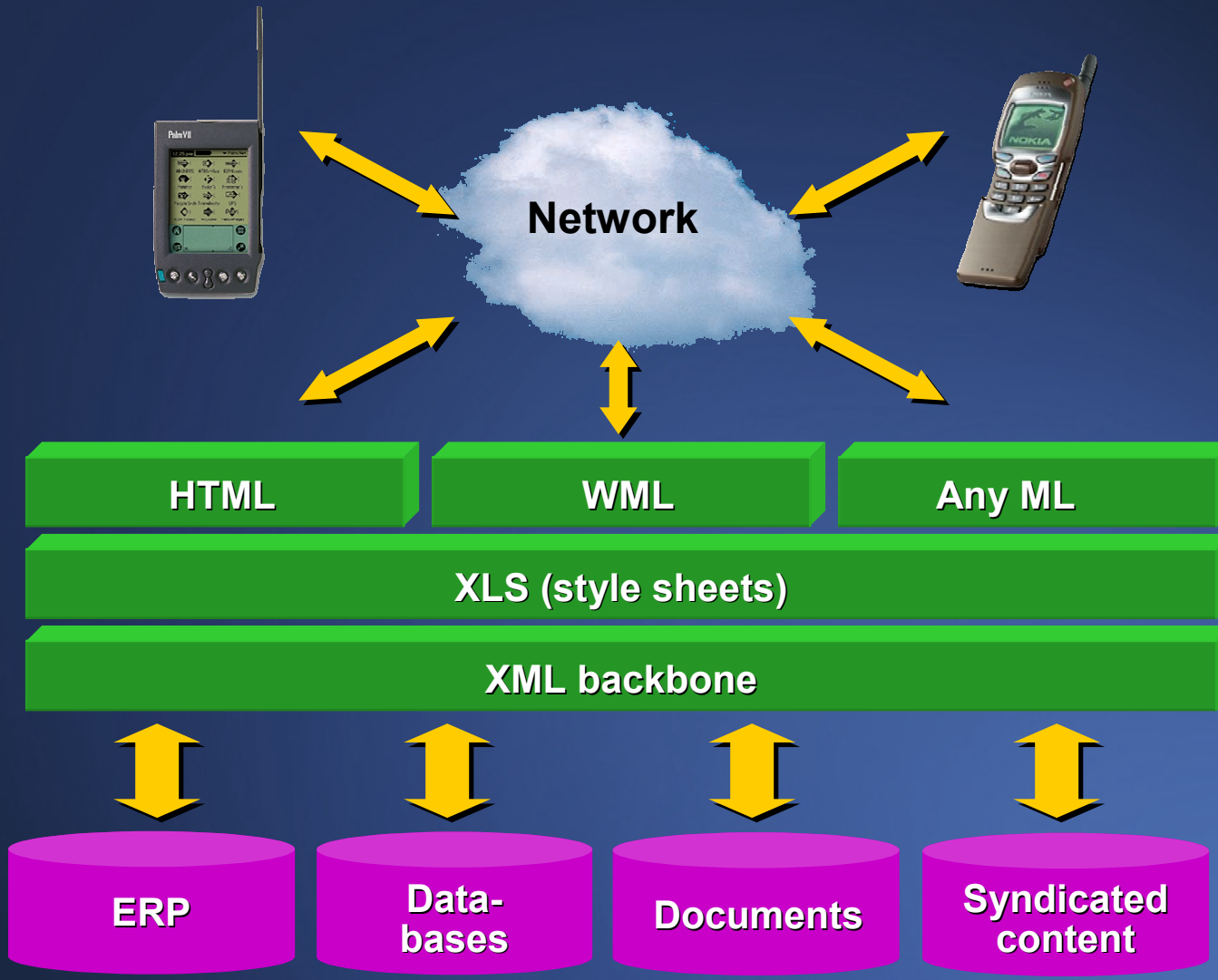
XML Primer (Cont.)

```
<HTML>
<H1>Car for sale</H1>
<P>Model : Pontiac Firebird, Build in 1993.
<P>Red color and in good condition. For those who
   care : 150 horse power.
<P>Contact : Rob Janson, 425 462 1999
</HTML>
```

XML Primer (Cont.)

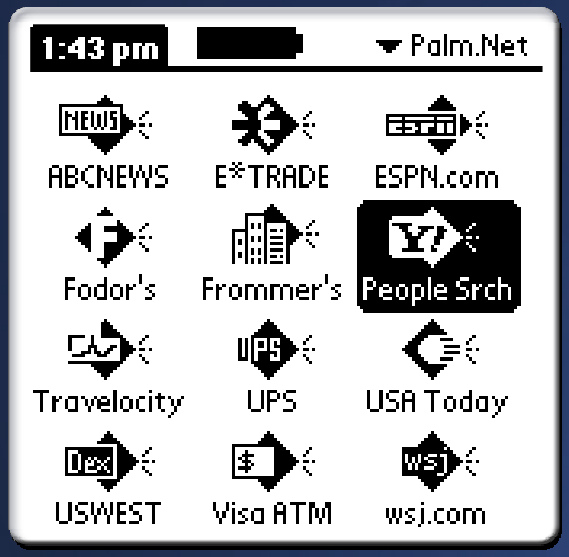
```
<Sale What="Car">
  <Model>
    <Company>Pontiac</Company>
    <Type>Firebird</Type>
    <Color>Red</Color><Horsepower = "150"/>
  </Model>
  <Contact>
    <Name>Rob Janson</Name>
    <Phone>425 462 1999</Phone>
  </Contact>
</Sale>
```

XML—Data Backbone



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Palm Web Clipping



Launcher



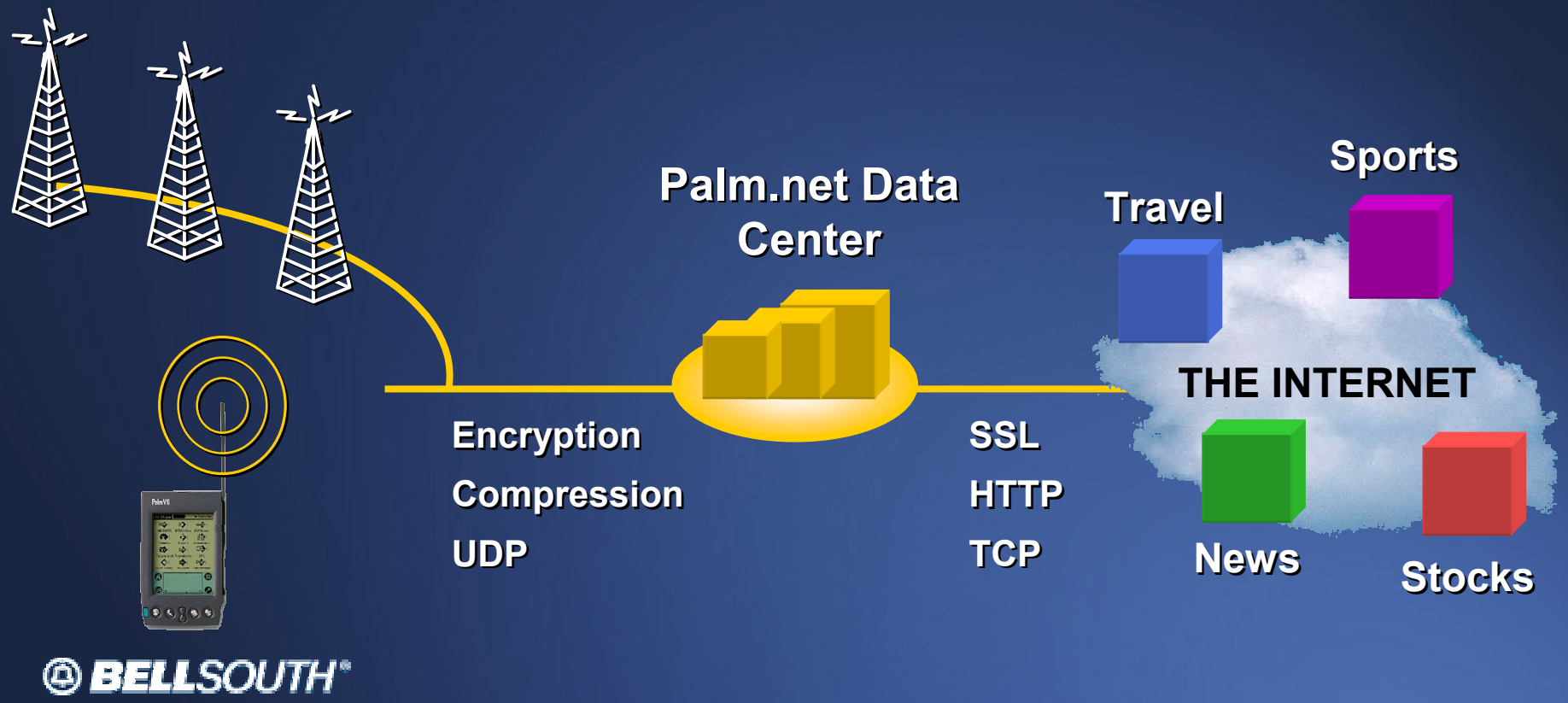
Query Application



Clipping

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Palm.net—Internet Access



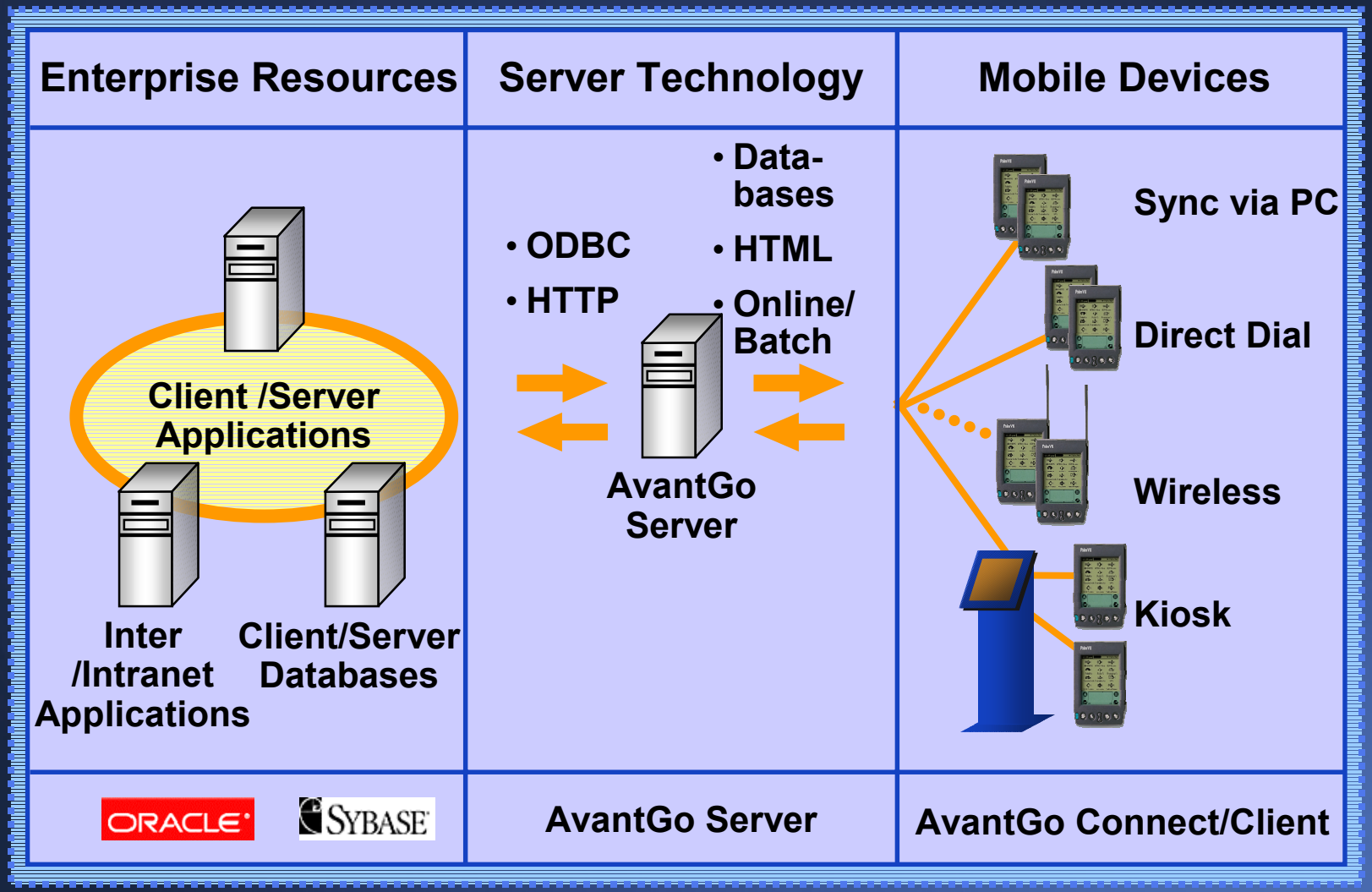
WAP Browsers for Palm

- WAPman by EdgeMatrix
- AU Browser by AU-Systems
- ThinkMotion Browser by ThinkersGroup
- WAP browser for Palm by MobileID
- Neomar Microbrowser
- kBrowser for Palm by 4thpass

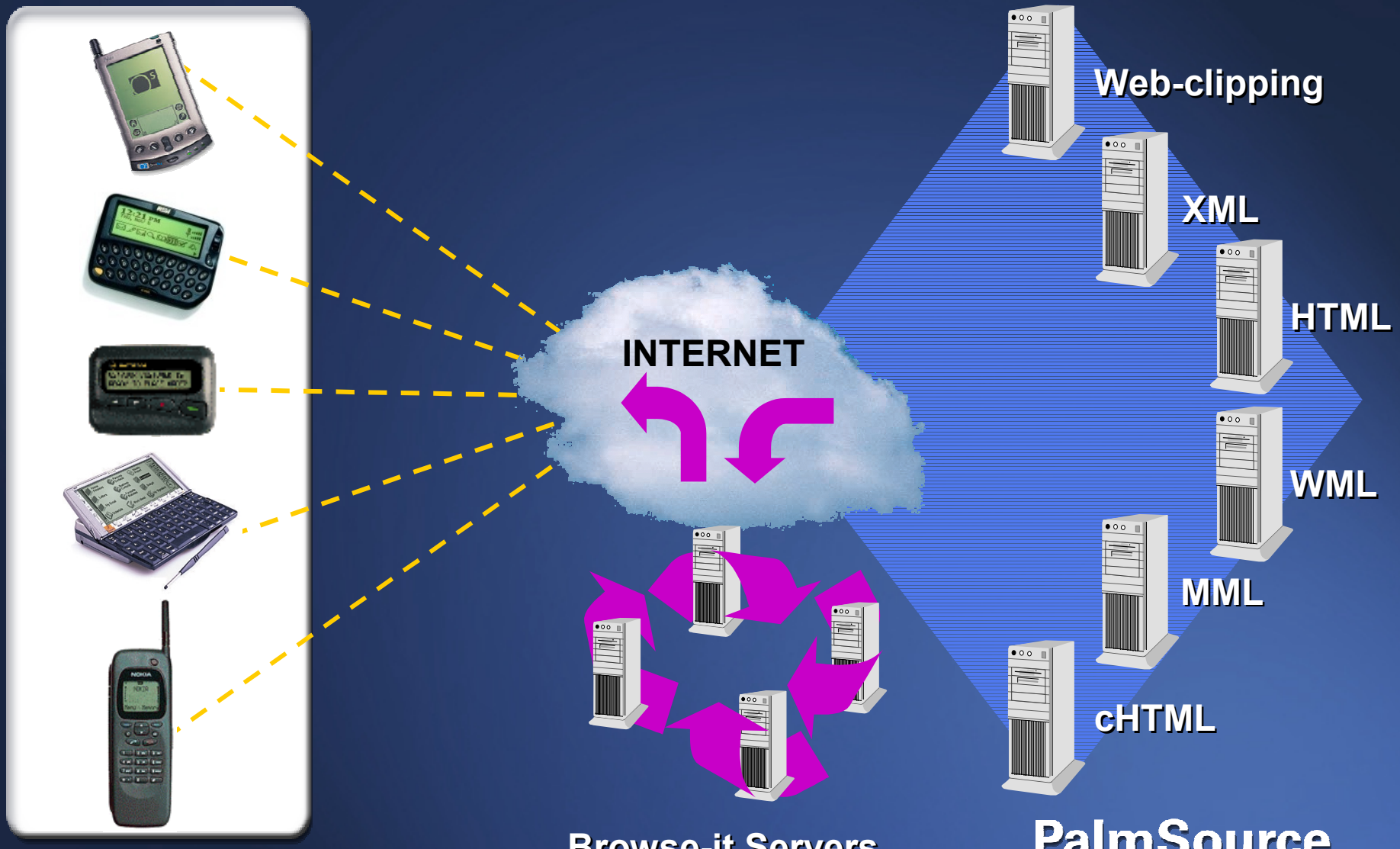
Proxi Servers vs. Transcoding

- Proxi Server
  - Takes content and conditions it for a companion client
  - Pros:
    - Tighter integration with handheld
    - Better offline browsing capabilities
  - Cons:
    - Proxi vendor must create custom client for each device
- Transcoder
  - Takes a markup language and converts it to a mobile/wireless markup language
  - Pros:
    - Can be accessed with standards compliant browser
    - Ensures better cross-platform support
  - Cons
    - Inferior offline browsing

Proxi Servers—AvantGo



Proxi Servers—Puma Browse-it



Transcoders/Gateways

- IBM Transcoding Publisher
- Oracle 9i Wireless Edition
- Phone.com UP.Link Server
- Nokia Activ Server
- Aether ScoutWeb
- Other providers
  - Bluestone, Thinkers Group, Infinite.com, Neomar, Everypath, 2Roam, DataChannel, Wireless Knowledge



Building Wireless Applications —Client/Server (non-Web)

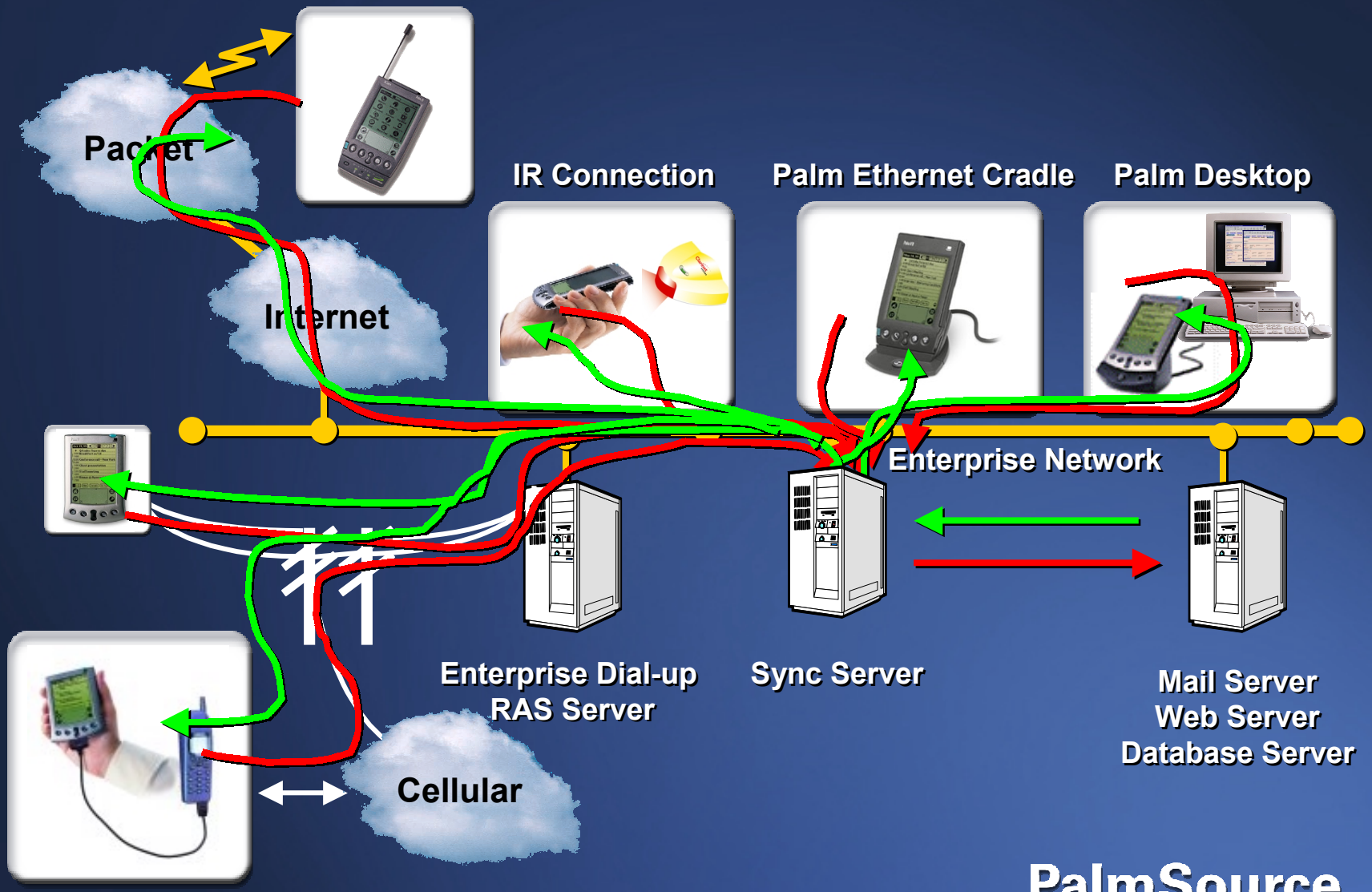
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Requirements Beyond Web

- Data synchronization
- Wireless email and groupware
- Device management
  - Inventory
  - Backup/restore
  - Application distribution
- Feature rich client applications
- Offline data viewing and processing
- Seamless data access both wireless and wireline

Synchronization Environment



Value of Server Synchronization

- Central management of handhelds
  - Application distribution
  - Inventory
  - Data backup/restore
- Control information access
  - Account name and password
  - User groups
- Remove PC as required component

Synchronization Servers

- Palm HotSync Server
  - Email, custom development
- Puma IntelliSync Anywhere, Sat Forms Server
  - Email, custom development
- IBM Mobile Connect/XTND Connect
  - Email, custom development
- Synchrologic
  - Database integration tools
- AvantGo
  - Email, web, database

Handheld Server—Real Time Access

- Why real time rather than synchronization?
  - Financial transactions require immediate closure
  - Back-end data too dynamic to allow offline transactions
  - Efficiency, convenience
- Aether Technologies AIM
  - Support for CDPD, Motient and Mobitex
- Centura Software eSNAPP
  - IP based



Third-Party Hosting/ Wireless Portals/ WASPs (Wireless Application Service Providers)

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WASP Benefits

- Single vendor for client hardware, network access and content
- Support for heterogeneous client environment and multiple wireless networks
- Value-add information services
- Quickly scalable

WASPs—Vertically Oriented

- Cresenda
  - Industry specific hosting and deployment services
  - Support for CDPD
- Aether Wireless Internet Services
  - Application development, client/device management, carrier services, customer care
  - Support for CDPD, cellular, Mobitex, Motient
- Broadbeam SystemsGO!
  - Managed hosting programs—pilot, deployment and ongoing management
  - Support for CDPD, cellular, Mobitex, Motient

WASPs—Broad-based Oriented

Provider	Services	Services
fusionOne	web, alerts, synchronization	cellular, CDPD
Go America	enterprise hosting, email, web	BSWD, CDPD
Intellisync.com	web, alerts, synchronization	CDPD, cellular
OmniSky	email, web, info services	CDPD
Palm.net	web, messaging	BSWD, CDPD, cellular
Yada Yada	email, web, info services	CDPD



Software Tools

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PalmVII/VIIx Applications for BSWD

- Web clipping applications
  - Work with HTML
  - Standard authoring tools
  - QAB—Query Application Builder
- PalmVII SDK
  - Create wireless applications in C,C++
  - Metrowerks Codewarrior
  - Access to INetLib

Palm Applications Based on TCP/IP

- For cellular, wireless IP and landline telephone based applications
- Based on Berkeley UNIX sockets API
- TCP/IP stack runs as a separate task in the Palm OS
- Support for PPP and SLIP

Broadbeam™ Tools for Palm

- SmartIP™
  - TCP sockets based interface
  - Enables client/server TCP/IP applications to run over IP and non-IP networks
- ExpressQ™
  - Store-and-forward messaging uses database based message queues
  - Pushes messages from server to client
  - Event driven architecture

Other Wireless Tools

- Certicom Security Builder
  - Tools to create encryption, digital signatures and key exchange mechanisms for Palm applications
- Aether Client SDK, Server SDK
  - Abstracts unique transport requirements of different networks
  - Build once, run on any network

Summary

- Network Considerations
  - Coverage, cost, performance
- Application Architecture
  - Web vs. non-Web, batch vs. real-time, cross-platform considerations, security
- Build your own vs. WASP
  - Cost, viability, level of service, responsiveness


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